Traditional Search & Rescue Collapses Across 4 Critical Voids

70% CANOPY CRASHES | 15-30m GEO ERROR | 2-3 HOURS DELAY | ₹40L+ AIRFRAMES

SUBSYSTEM A // GNC

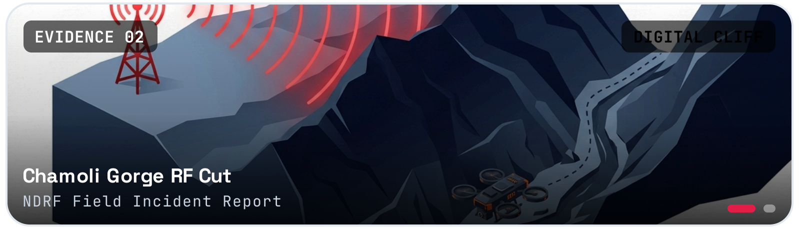
Canopy GPS Drift & Crashes

CORE FAILURE MECHANISM:
Dense foliage blocks satellite GNSS signals

OPERATIONAL IMPACT:
Commercial drones lose position lock, drifting into tree canopies and causing rotor loss.

70% Loss	0 Hold
Canopy Sorties	Non-GPS Stability

TARGET: PX4 & 3D ORCA — LEAD: NIKHIL



SUBSYSTEM B // COMMS

Mountain Ravine RF Blackout

CORE FAILURE MECHANISM:
Ridgelines sever direct line-of-sight RF

OPERATIONAL IMPACT:
Conventional H.264 digital video completely cuts out below 5dB SNR, causing total blindness.

<5dB SNR	0 Relay
Video Blackout	Single-Drone Link

TARGET: DEEP JSCC MESH — LEAD: NIKHIL

SUBSYSTEM C // VISION

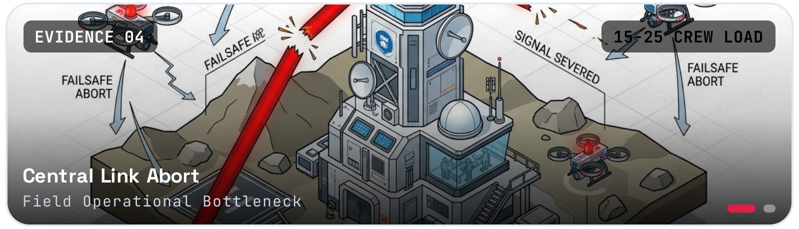
Flat-Earth Elevation Drift

CORE FAILURE MECHANISM:
2D raycasts assume flat zero-elevation ground

OPERATIONAL IMPACT:
Sloping terrains produce 15-30m coordinate errors, routing ground teams to empty ravines.

15-30m	35%
Location Drift	False Alarm Rate

TARGET: 3D DEM RAYCAST — LEAD: VEDANTH



SUBSYSTEM D // C2 GCS

Central Pilot Bottleneck

CORE FAILURE MECHANISM:
1-pilot-per-drone manual radio control

OPERATIONAL IMPACT:
Requires 15-25 personnel and 45-90 min setup; sortie aborts if the single pilot link drops.

2-3 Hrs	₹12.5L
Search Time / mi²	Cost / Deployment

TARGET: WEBGPU ATAK GCS — LEAD: SIVA

FIELD EVIDENCE	AUDIT FINDING	COST PENALTY	SURVIVOR RISK
Wayanad Landslide Audit	70% RF/GPS Disconnect	₹40L Military Airframes	412% Disaster Surge